

assign_functional_type_clusters.R

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Assign Functional-Type Clusters

Description

Assigns taxa into functional types (FTs) using a pre-computed Gower dissimilarity matrix, a pre-fitted hierarchical clustering object, and a predetermined number of groups ‘functional_type_group_count’. The number of groups is expected to have been chosen upstream via ‘select_functional_type_group_count()’.

Usage

```
assign_functional_type_clusters(  
  data_trait_table,  
  trait_dissimilarity,  
  hierarchical_clustering,  
  functional_type_group_count,  
  taxon_column = "taxon_name",  
  verbose = TRUE  
)
```

Arguments

data_trait_table
A data frame with one row per taxon. Must contain a character column (identified by ‘taxon_column’) holding taxon names. Must have at least 4 rows and ‘nrow(data_trait_table)’ must be strictly greater than ‘functional_type_group_count’.

trait_dissimilarity
A ‘dist’ object produced by ‘compute_trait_dissimilarity()’. Must inherit class “dist”. All distance values must be finite.

hierarchical_clustering
An ‘hclust’ object produced by ‘fit_hierarchical_clustering()’. Must inherit class “hclust”.

<code>functional_type_group_count</code>	A single positive integer giving the number of functional-type groups to cut the dendrogram at. Must be at least 2 and less than <code>nrow(data_trait_table)</code> .
<code>taxon_column</code>	A single character string naming the column in <code>'data_trait_table'</code> that contains taxon names. Default: <code>"taxon_name"</code> .
<code>verbose</code>	A single logical. If <code>'TRUE'</code> (default), prints the chosen <code>'functional_type_group_count'</code> via <code>'cli::cli_inform()'</code> .

Details

****Algorithm**:**

1. Cut the dendrogram at `'functional_type_group_count'` via `'stats::cutree()'`.
2. Compute per-taxon silhouette widths via `'cluster::silhouette()'`.
3. Return a tibble of taxon names, cluster assignments, and silhouette widths.

Distance computation (Inf -> NA, daisy, NaN -> 1.0) and hierarchical clustering are handled upstream by `'compute_trait_dissimilarity()'` and `'fit_hierarchical_clustering()'` respectively. FT-groups selection is handled upstream by `'select_functional_type_group_count()'`. If `'nrow(data_trait_table) < 4'` or `'functional_type_group_count >= nrow(data_trait_table)'`, the function aborts with an informative error.

Value

A tibble with columns:

taxon_name Taxon names (same values as `'data_trait_table[[taxon_column]]'`). Column name is always `"taxon_name"` regardless of `'taxon_column'`.

functional_type Integer label (1..`'functional_type_group_count'`) giving the functional-type cluster assignment for each taxon.

silhouette_width Per-taxon silhouette width for the chosen `'functional_type_group_count'` solution.

See Also

`[compute_trait_dissimilarity()]`, `[fit_hierarchical_clustering()]`, `[select_functional_type_group_count()]`

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